The Landscape of Short Tandem Repeat Variation

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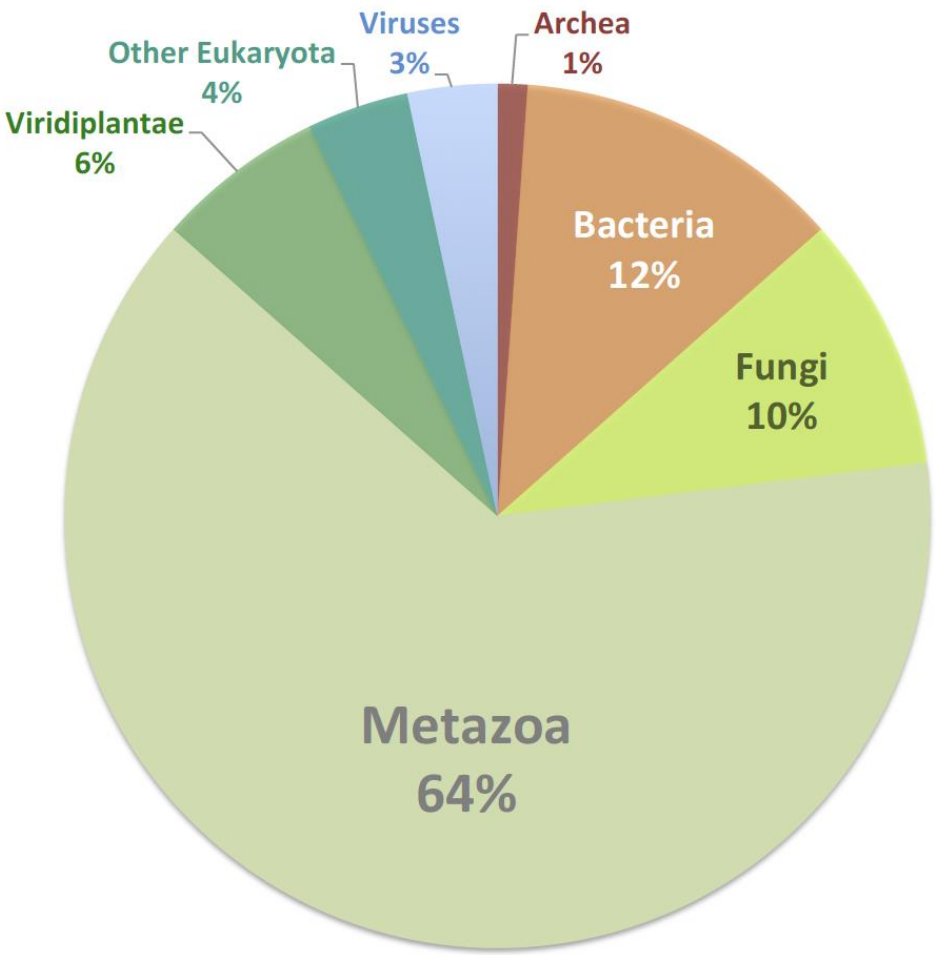
Tandem repeats (TRs)

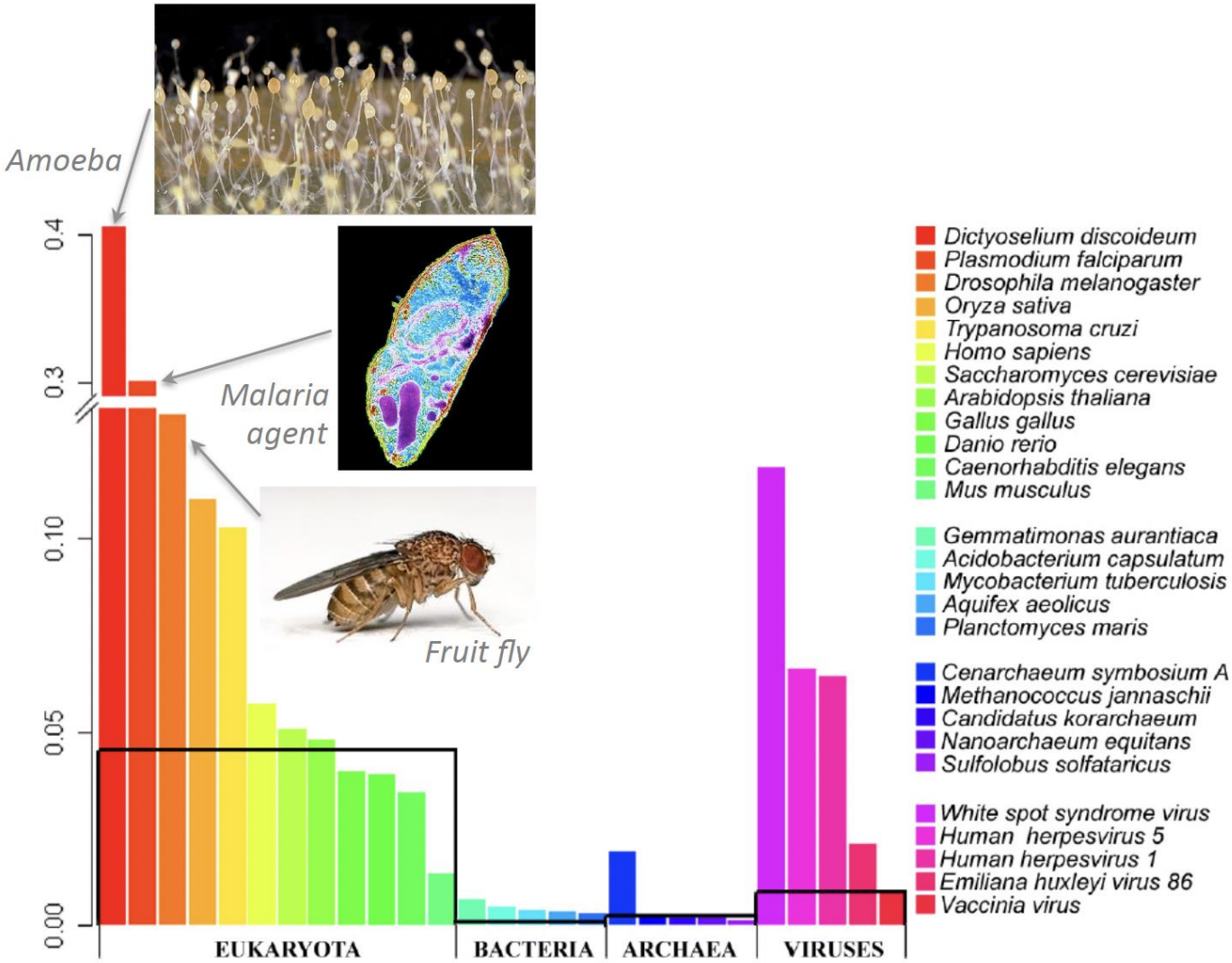
- Abundant: > 60 % of human proteome (Schaper et al 2012)
- Enhanced binding properties
- Often associated with disease / pathogenicity

Protein TRs are found across the tree of life



Distribution in SwissProt (based on Andrade et al. 2001)

Frequency of proteins with homorepeats: NCBI RefSeq



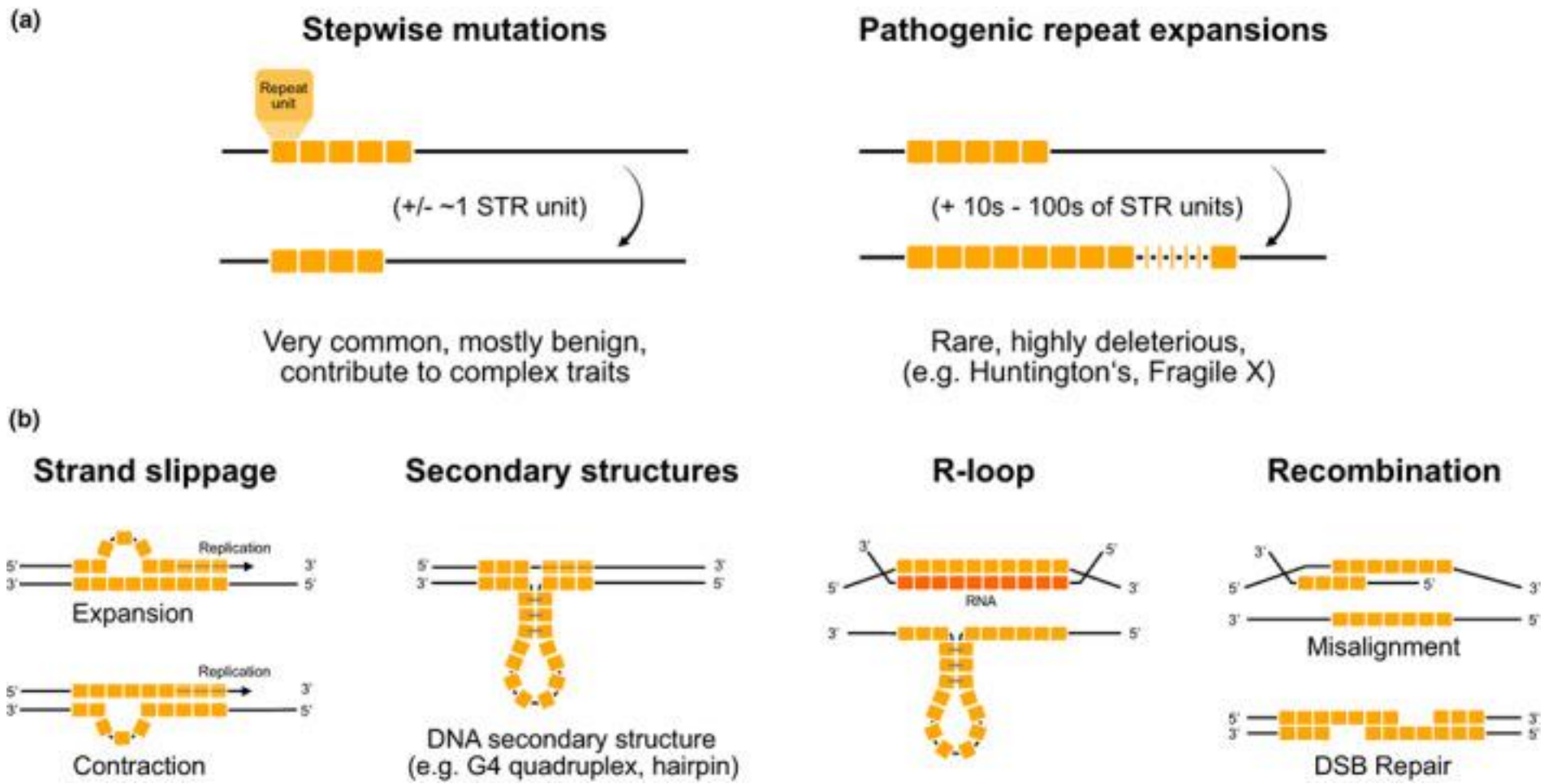
from Jorda and Kajava 2010

STRs: Short Tandem Repeats

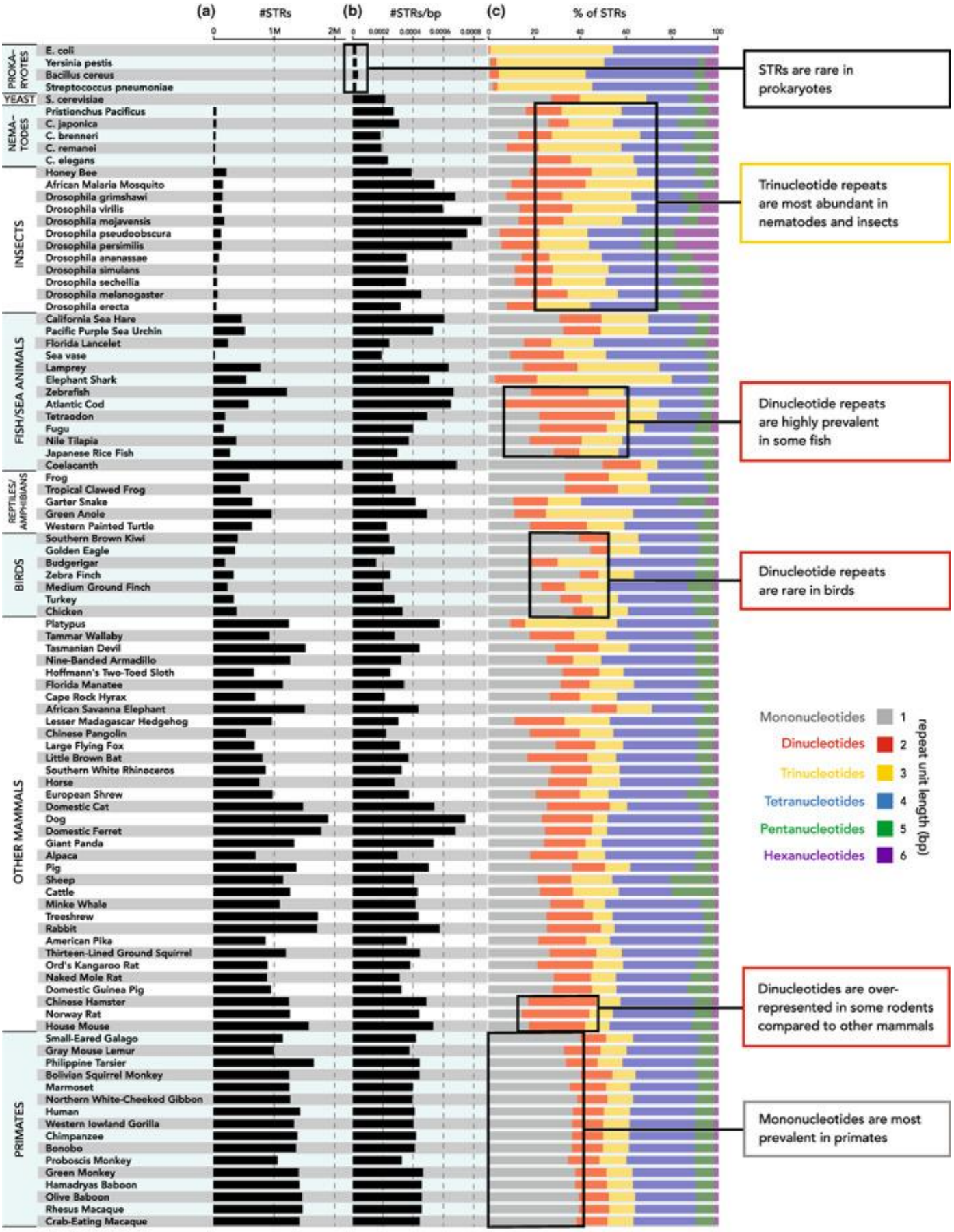
STRs = repeats of 1-6 bp DNA units (*a.k.a.* microsatellites)
Mutations rates > 10’000 x higher than point mutations

```
5’ ... C G A C A T A T A T A T A T A T A T A T A T A T G A T C ...
       | | | | | | | | | | | | | | | | | | | | | | | | |
3’ ... G C T G T A T A T A T A T A T A T A T A T A T A C T A G ...
```

Majority of STR changes are stepwise



Verbiest *et al*., (2023). Mutation & selection processes regulating STRs give rise to genetic & phenotypic diversity across species. *J Evol Biol*



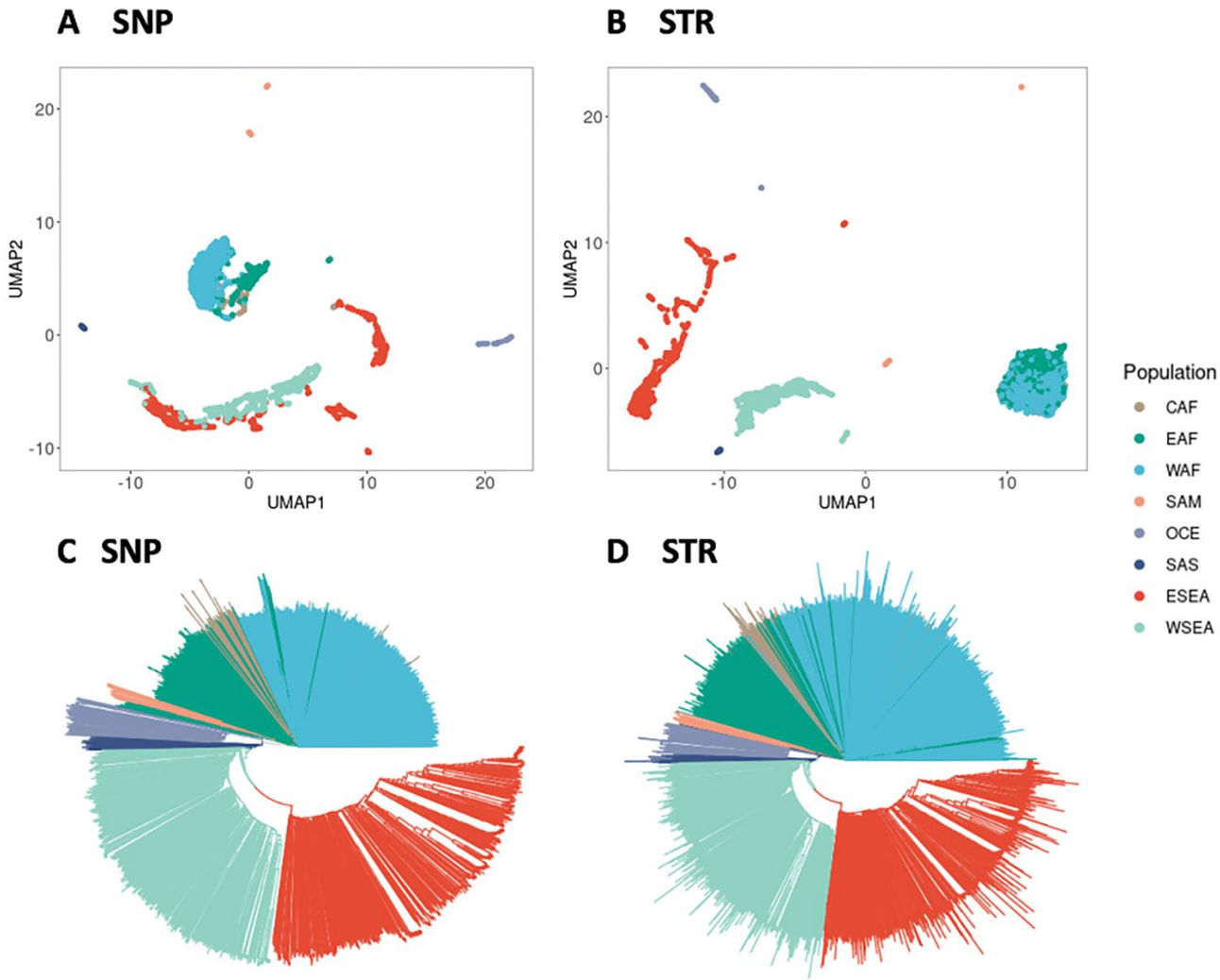
Verbiest et al 2023 J Evol Biol

STRs are informative about population structure

Fig 3. Population structure analysis of the 3047 *P. falciparum* samp les of SNP and STR data.
UMAP clustering of the top five principal components of the SNP (A) and STR data (B),

Neighbor joining tree based on the SNPs (C) and STR data (D).

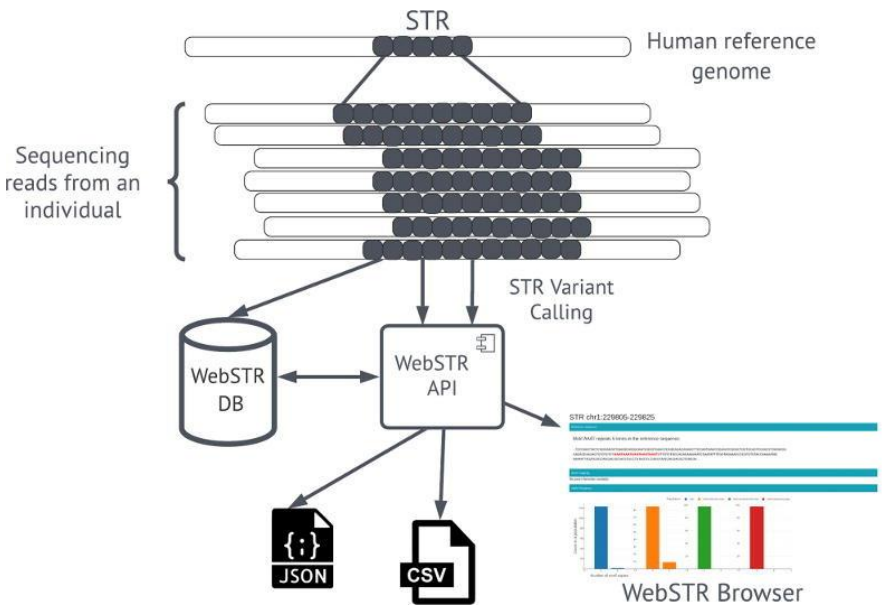
Colours representing the eight different populations.



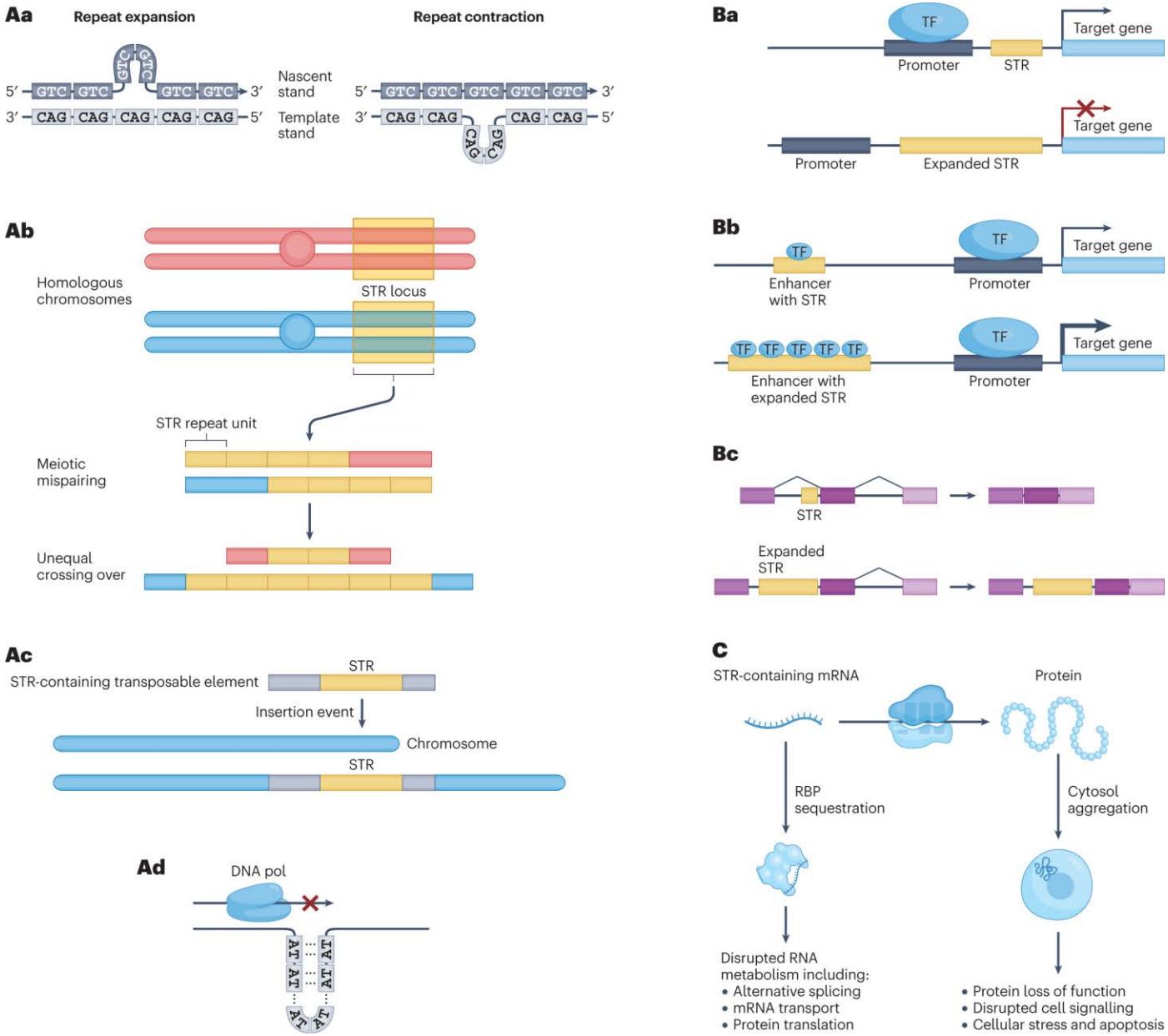
Han et al. 2022. PloS Genetics, https://doi.org/10.1371/journal.pgen.1009604

WebSTR: A Population-wide Database of Short Tandem Repeat Variation in Humans

Oxana (Sachenkova) Lundström [1,2,3], Max Adriaan Verbiest [3,4,5], Feifei Xia [3,4,5], Helyaneh Ziaei Jam [7], Inti Zlobec [6], Maria Anisimova [3,4]* and Melissa Gymrek [7,8,*]

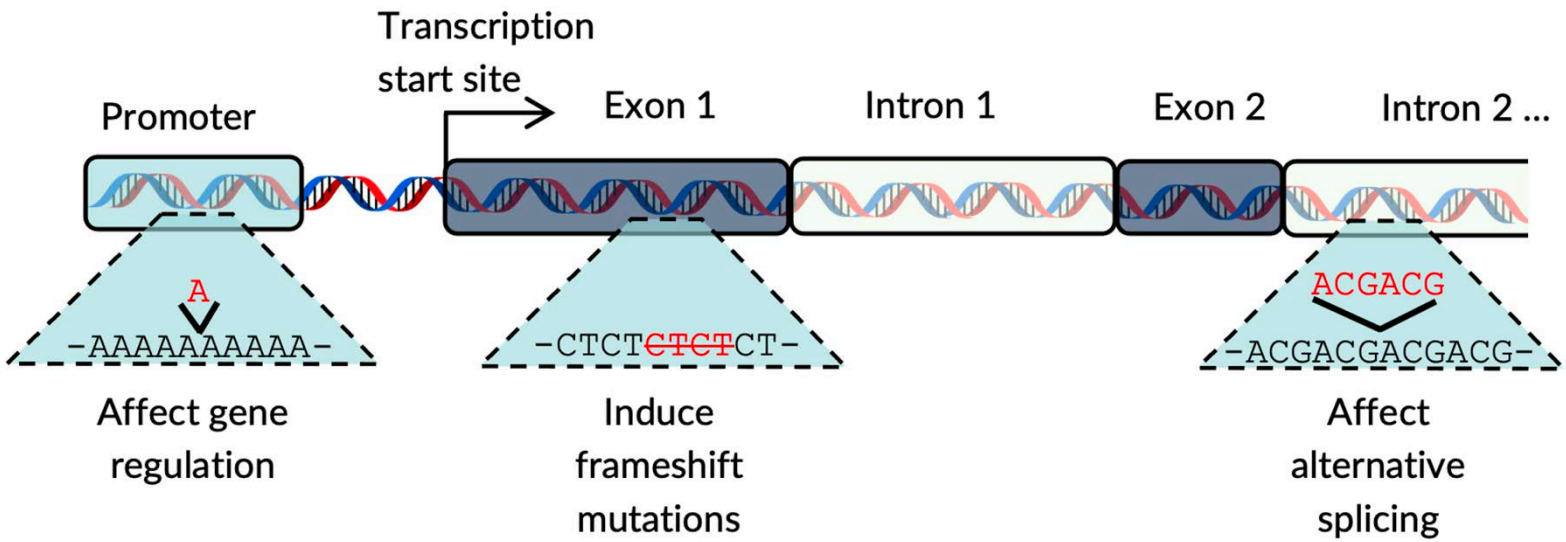


Causes & consequences of STR variation



Tanudisastro et al. (2024) *Nature Rev Genetics*

STRs can regulate gene expression

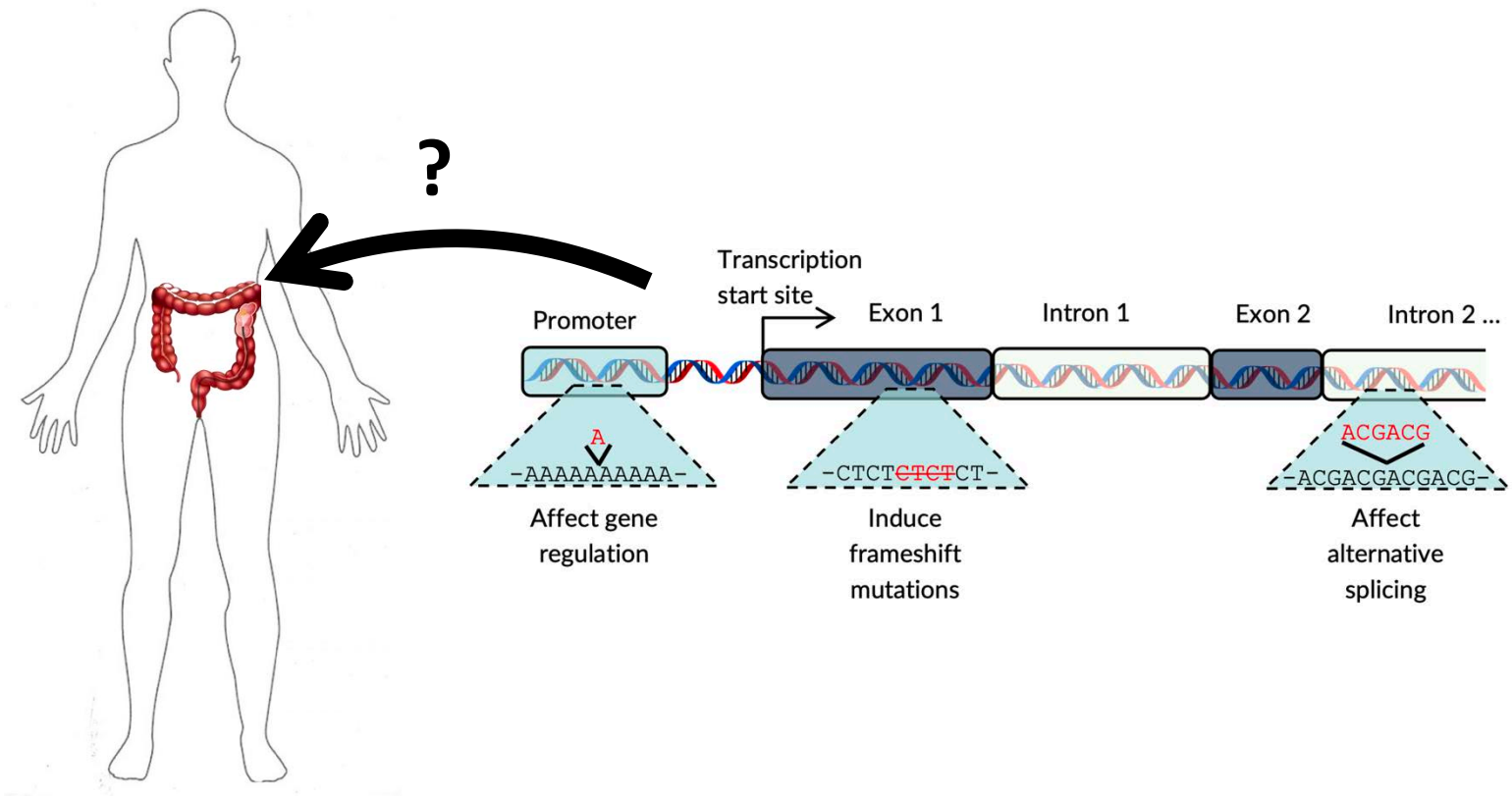


Such STRs are known as *eSTRs* – expression STRs

Fotsing, F. *et al*., (2019), The impact of short tandem repeat variation on gene expression, *Nature Genetics*

Role of eSTRs in colorectal cancer (CRC) is unclear

Our goal: identify eSTRs in CRC

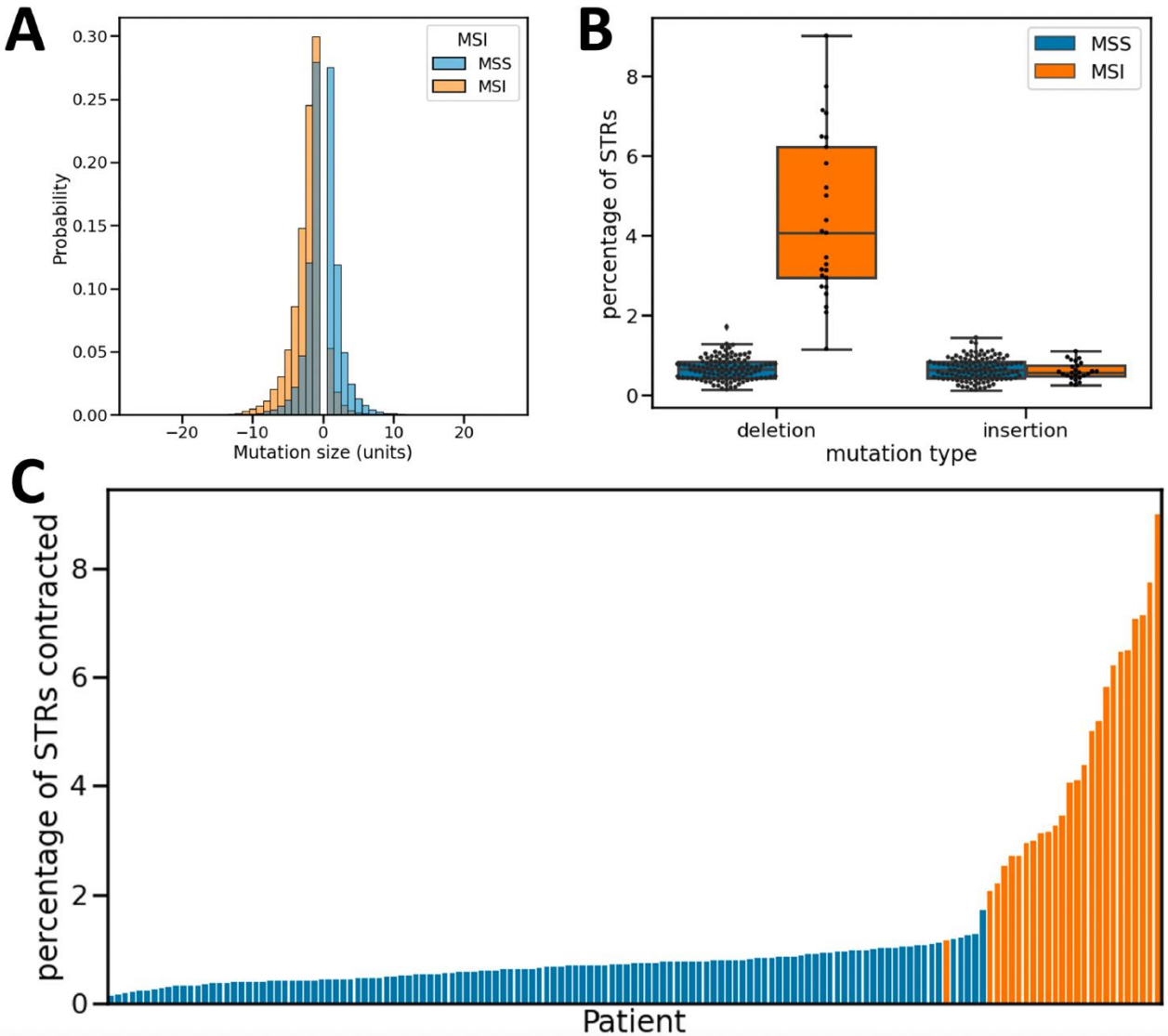


Verbiest et al (2024), Scientific Reports

Detection of eSTRs in CRC

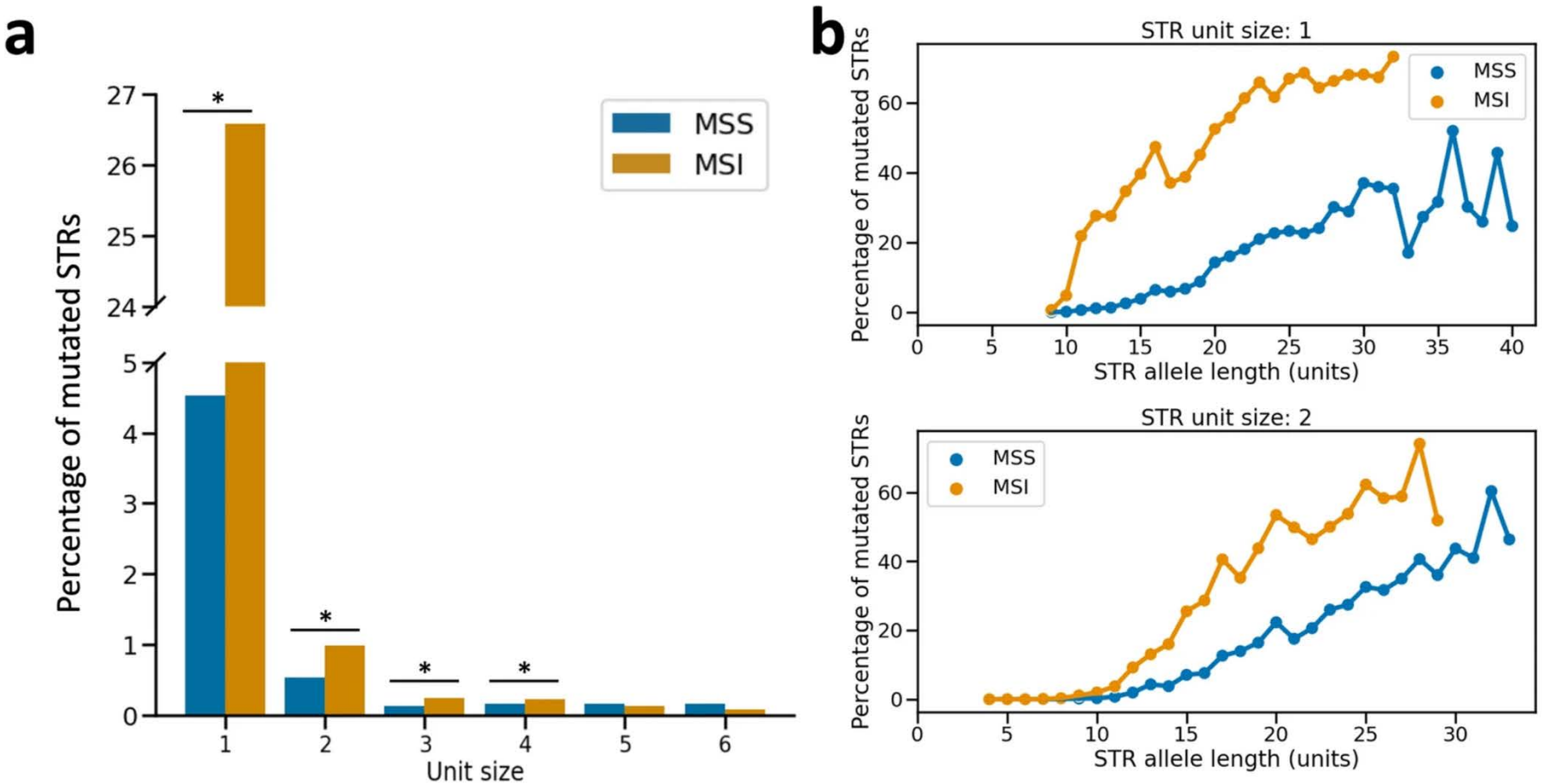
- Data: sequencing & gene expression for 311 primary tumour samples from TCGA
- Tested for linear associations between repeat copy numbers and gene expression for pairs of STRs and genes
- Identified significant relationships for 1289 STR-gene pairs (eSTRs)
- 71 of these genes are COSMIC cancer genes

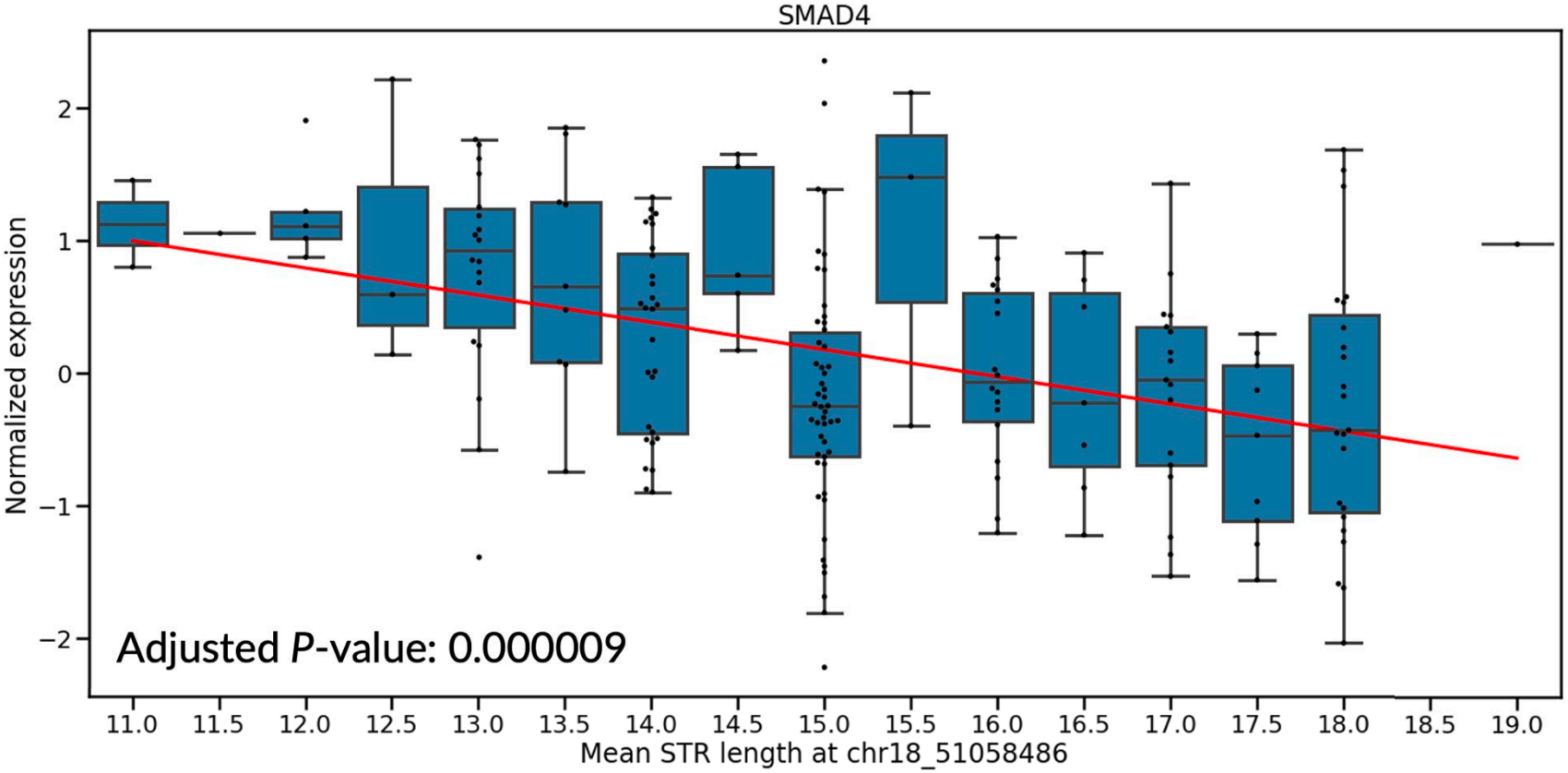
CRC sequencing data from TCGA*



* Cancer Genome Atlas Network. (2012). Comprehensive molecular characterization of human colon & rectal cancer. *Nature*

Mutated STRs by unit size

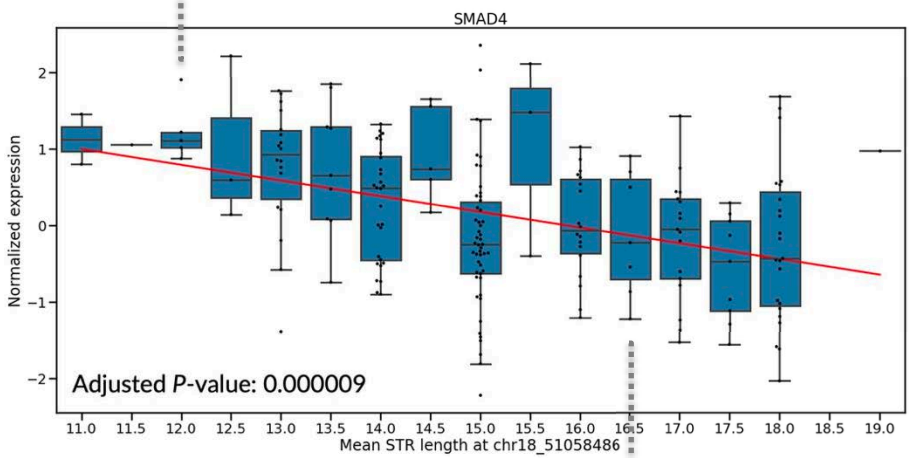


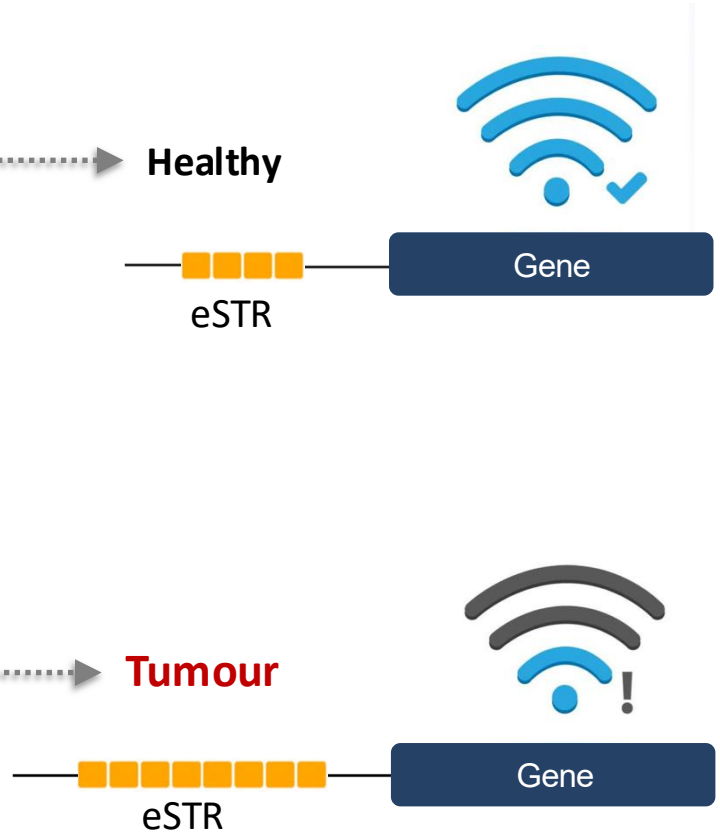


The role of TGF-β/SMAD4 signaling in cancer

Ming Zhao[1,2], Lopa Mishra[3]✉, and Chu-Xia Deng[1]✉

Example from Verbiest et al (2024): SMAD4 eSTR lenth vs. expression





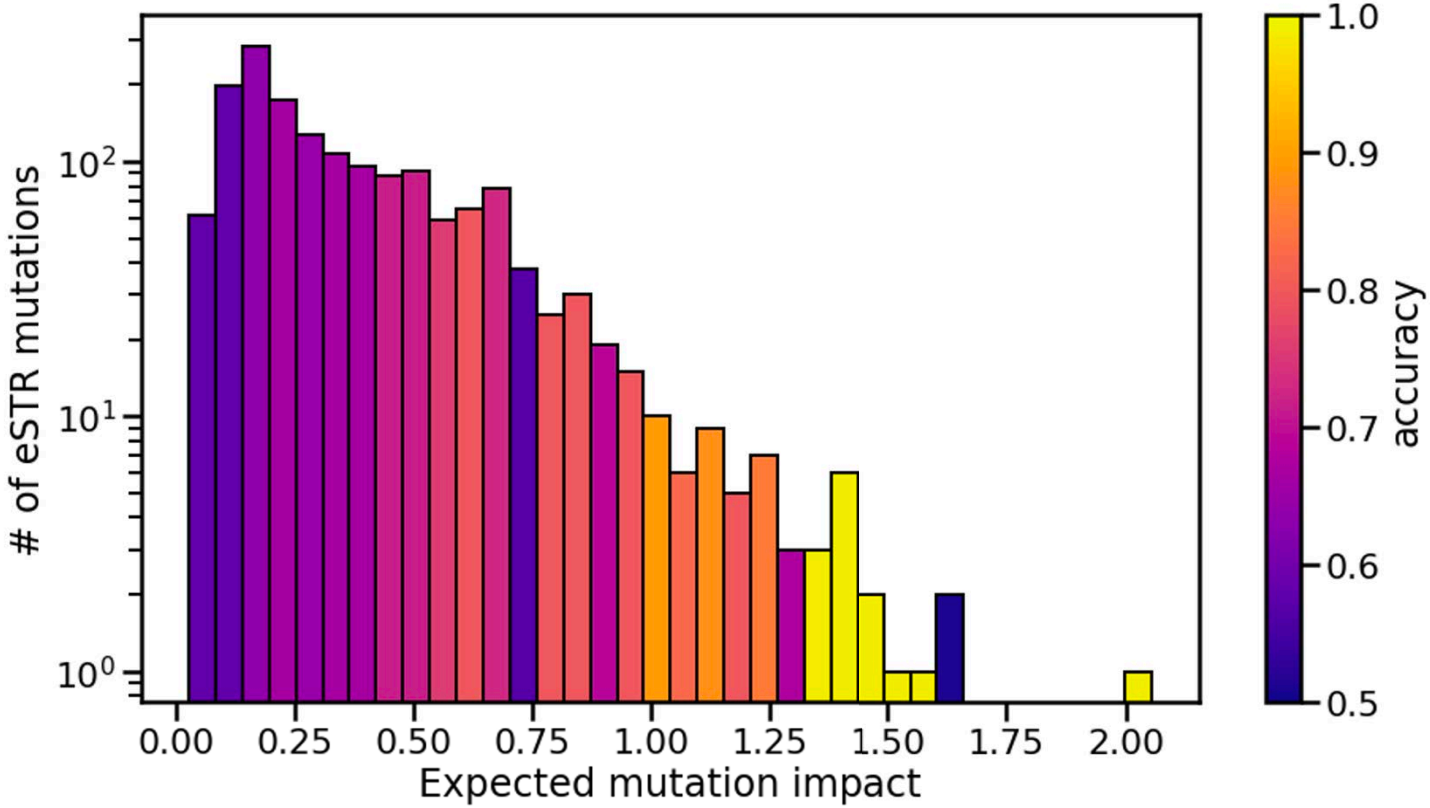
Predicting gene expression changes using eSTRs

- Hypothesis: mutations at eSTRs should cause gene expression changes
- To investigate this, we looked for eSTR mutations in a separate set of CRC tumours with patient-matched healthy samples

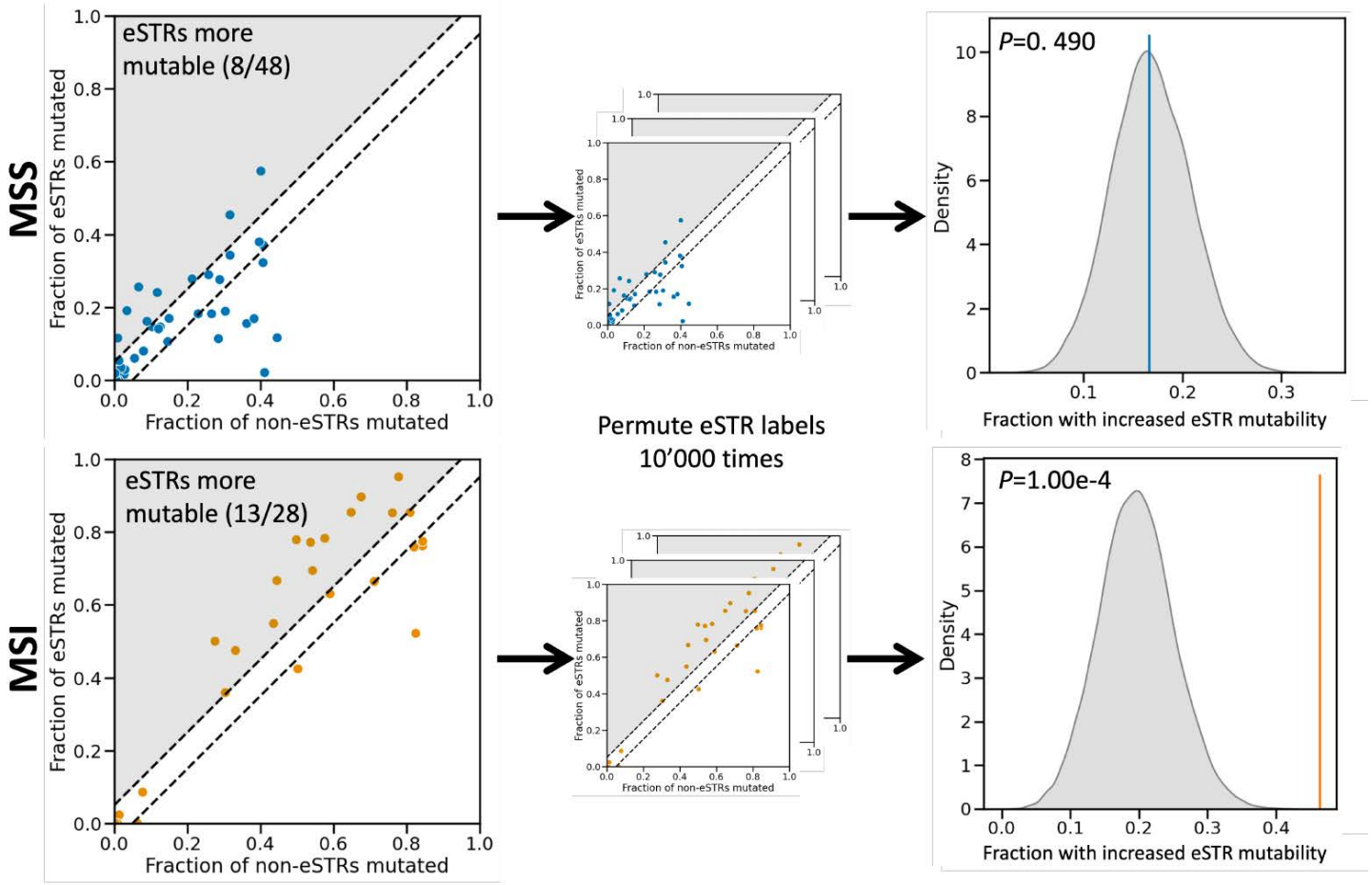
eSTRs predict gene expression changes

- Identified 1584 eSTR mutations in the validation set
- For 66.7% of mutations, the eSTR association correctly predicted whether gene expression would increase or decrease (sig. better than random models; permutation test, $P \ll 0.05$)
- Mutations at eSTRs with strong associations, or mutations resulting in large copy number differences, are expected to have a larger impact on expression
- Prediction accuracy was higher for eSTR mutations with higher expected impact

Higher prediction accuracy for eSTRs with higher expected impact



Compared to non-eSTRs, eSTRs are more mutable in MSI tumours



Summary

- STR mutations regulate gene expression in colorectal cancer
- Is this a mechanism for tumours to reprogram their phenotypes?
- eSTRs are more mutable in MSI tumours
- Future studies of eSTRs in other cohorts could increase our understanding of cancer(s), leading to novel targets for therapy
- Data available from *WebSTR*

eSTRs are more likely to be associated with DNA methylation

OXFORD

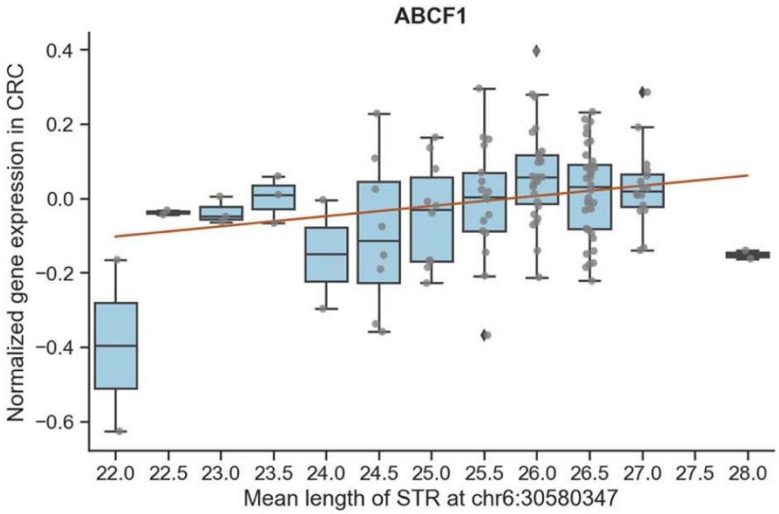
Briefings in Bioinformatics, 2025, **26(3)**, bbaf219

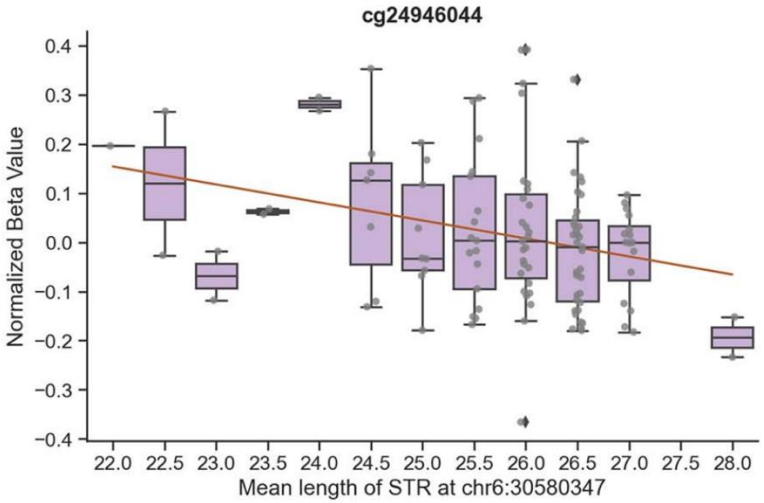
https://doi.org/10.1093/bib/bbaf219

Case Study

Multicancer analyses of short tandem repeat variations reveal shared gene regulatory mechanisms

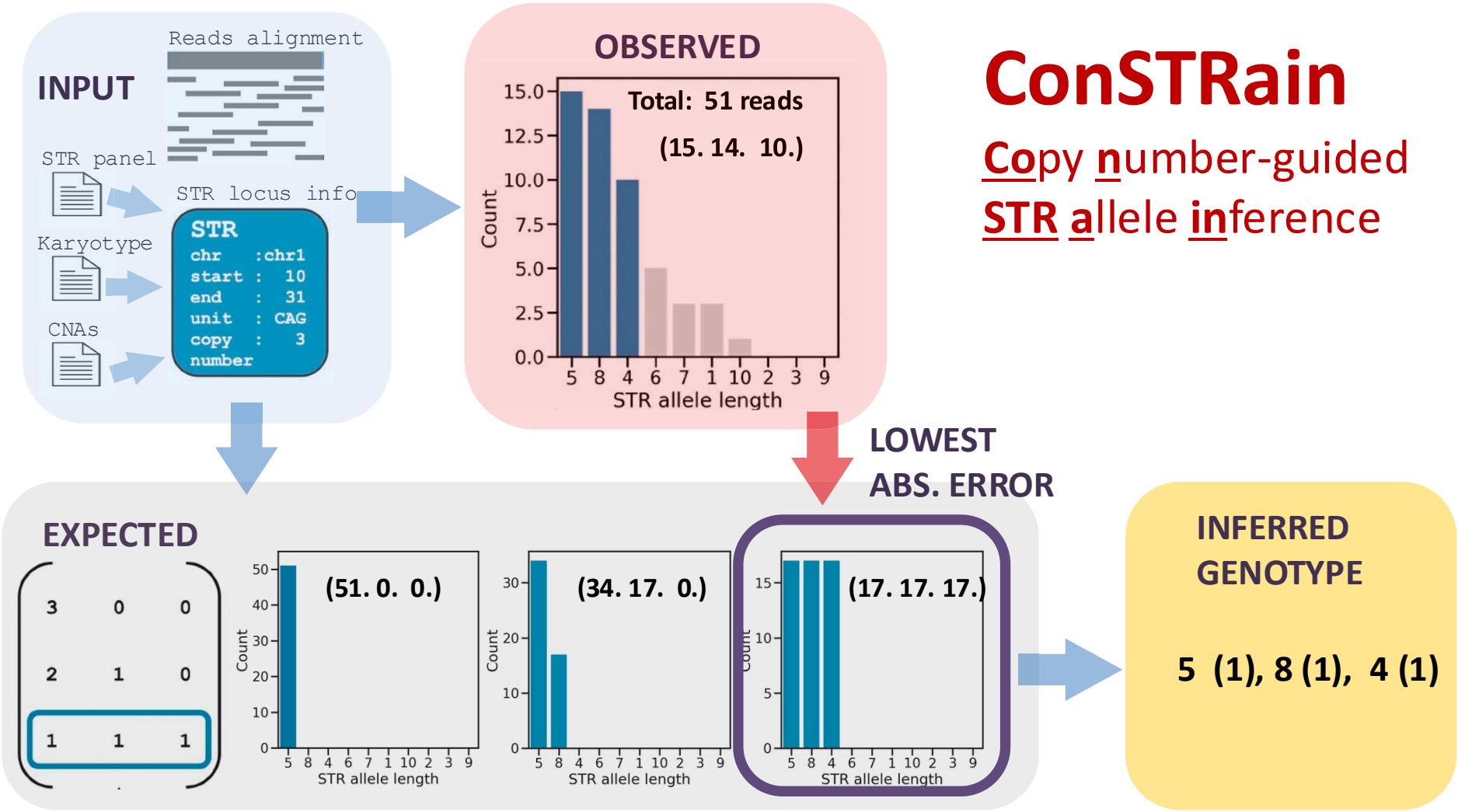
Feifei Xia [1,2,3], Max Adriaan Verbiest [1,2,3], Oxana Lundström [4], Tugce Bilgin Sonay [1,3], Michael Baudis [2,3], Maria Anisimova [1,3,*]

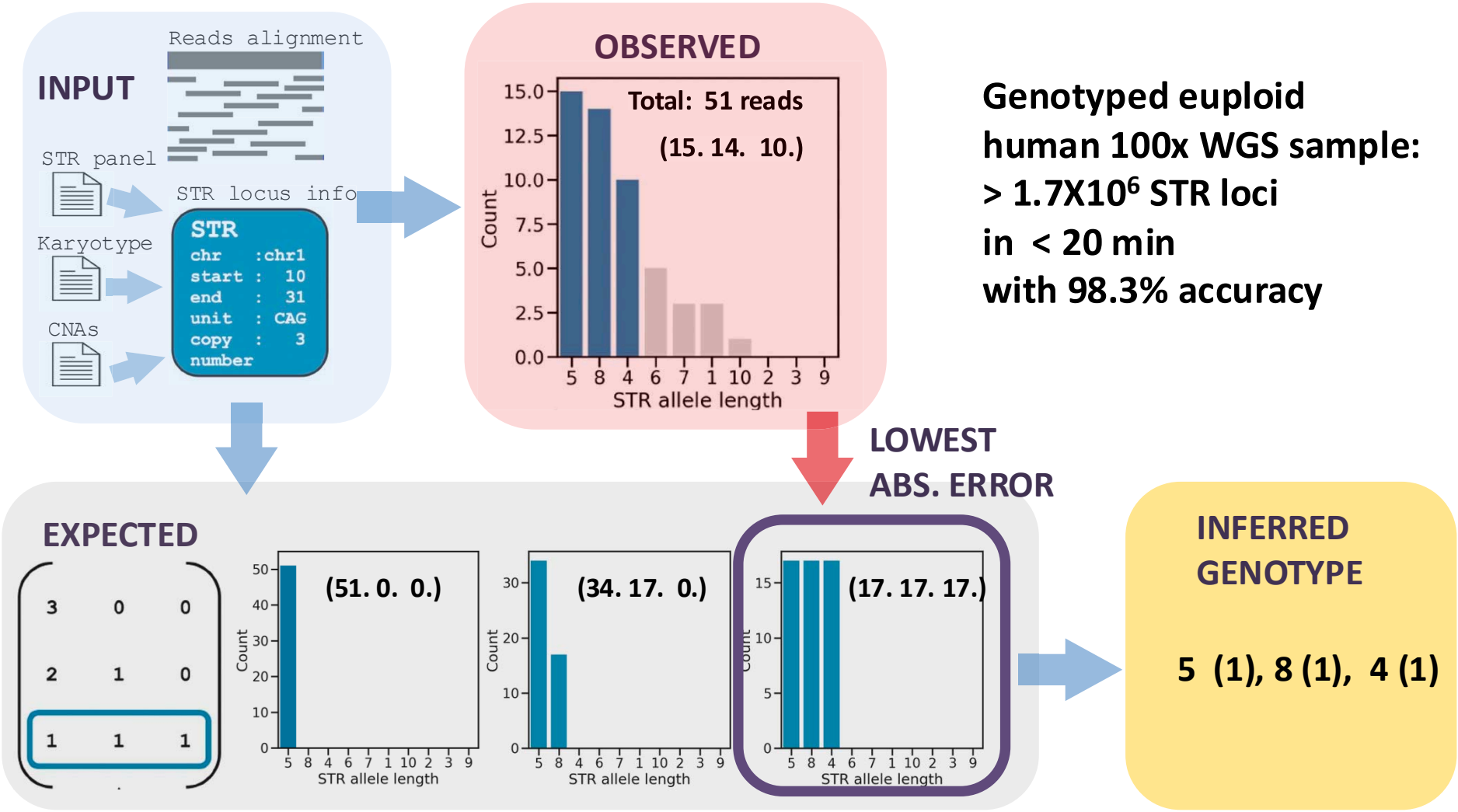




ConSTRain

Copy **n**umber-guided **STR** **a**llele **in**ference

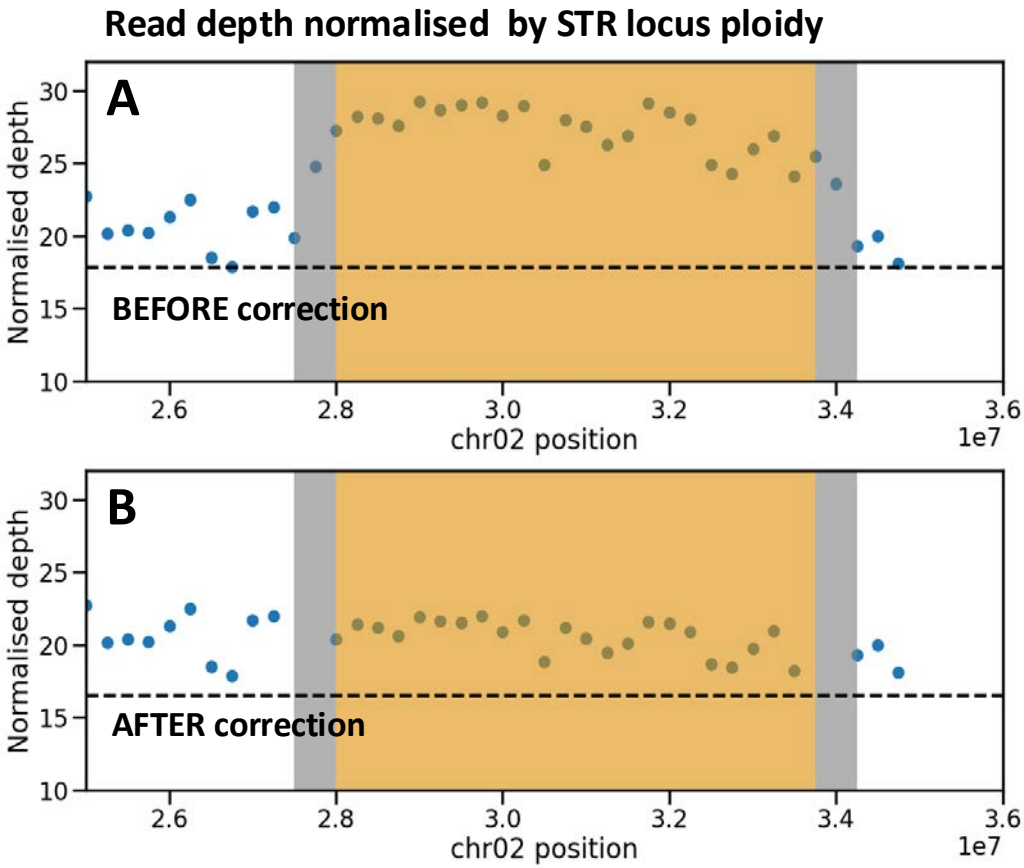




Genotyped euploid human 100x WGS sample:
> 1.7×10^6 STR loci
in < 20 min
with 98.3% accuracy

ConSTRain unlocks STR genotyping in polyploids

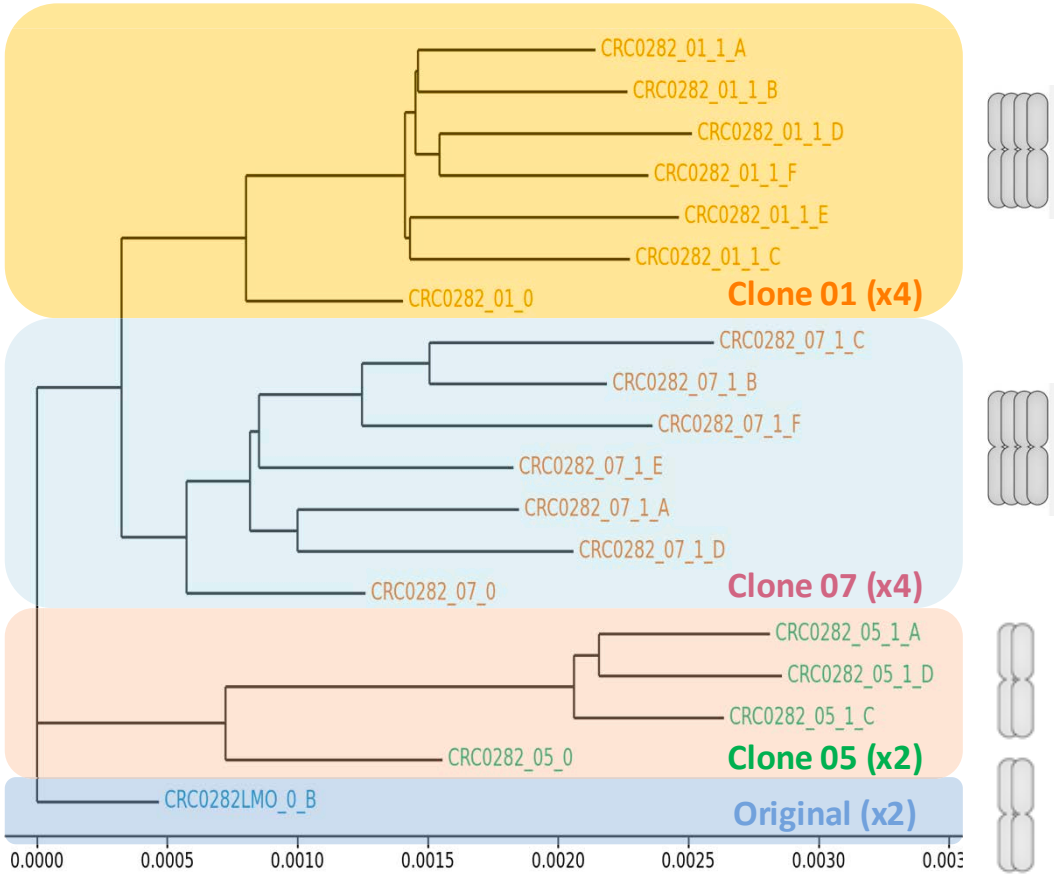
ConSTRain corrects for a large **genomic duplication on chr2** in a **triploid** Dwarf Cavendish banana genome



Musa acuminata

ConSTRain enables STR genotyping in cancer

Evolutionary tree inferred from STR genotypes for a **whole genome duplicated colorectal cancer tumoroid**



ConSTRain is flexible, fast, accurate STR variant caller accounting for ploidy-altering events

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Genotyping short tandem repeats across copy number alterations, aneuploidies, and polyploid organisms

Max A. Verbiest, Elena Grassi, Andrea Bertotti & Maria Anisimova

Thank you!

Swiss National Science Foundation

Universität Zürich[UZH]

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